

Ishaan Fartyal

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Final-year BSc Mechanical Engineering student at TU/e with a focus on engineering analysis, data interpretation, and quantitative modelling for strategic business decision-making. Experienced in Python-based analytics, operational and financial modelling, and cost analysis, applying data-driven insights to technology-driven businesses.

EDUCATION

BSc Mechanical Engineering — Eindhoven University of Technology (TU/e) Expected Nov 2026
Product Development & Entrepreneurship Elective Line (8 | 8 | 8) Eindhoven, Netherlands
Relevant Coursework: Dynamics of Energy Systems (9) | Solid Mechanics (8) | Multipled Robot (8)
Bachelor's Thesis: Determination of a Transition Plane in a Gas Atomization Simulation (Thermo-Fluids Engineering)

IB Diploma (High School) — K. International School in Tokyo (KIST) Jun 2021
43/45 | Physics HL 7 | Mathematics AA HL 7 | Economics SL 7 Tokyo, Japan

EXPERIENCE

Cost Engineer — University Racing Eindhoven (URE) Sep 2025 – Aug 2026
Formula Student engineering team Eindhoven, Netherlands

- Developed and maintained detailed Bills of Materials (BOM) and Costed-Carbonized Bills of Materials (CCBOM) of a race car to evaluate component costs and lifecycle emissions across manufacturing and distribution.
- Built an Excel and MATLAB cost model comparing prototype and mass-production scenarios for the inverter, evaluating in-house manufacturing and assembly versus outsourcing across production volumes to identify cost break-even points and sourcing implications.
- Assessed cost and carbon implications of component and manufacturing choices, translating engineering decisions into financial and sustainability trade-offs.
- Contributed the cost analysis that the team presented at Formula Student Czech Republic (11th of 52 in the Cost event) and Formula Student Germany.

Mechanical Design Engineer — University Racing Eindhoven (URE) Sep 2024 – Aug 2025
Formula Student engineering team Eindhoven, Netherlands

- Designed an in-wheel motor assembly for a race car using Siemens NX, translating design requirements, preferences, and constraints into a detailed CAD assembly.
- Evaluated packaging, manufacturability and integration constraints with vehicle sub-teams, iterating the design around technical and production limitations.

SELECTED WORK [↗](#)

Airbus A320 Ramp-Up Analysis [↗](#) Oct 2026
Independent Project

- Built a Python model of Airbus's A320 Family ramp-up to 75 aircraft a month, projecting yearly deliveries and cash flows to 2035 under limits on factory capacity, supplier availability and demand.
- Valued the ramp-up at approximately €1.0bn NPV over ten years and found it breaks even only if suppliers support about 82% of the plan, with each point of availability worth €0.33bn.
- Recommended timing investment to supplier readiness rather than a fixed schedule, as a two-year ramp-up delay turns NPV negative.

ASML Technology Strategy [↗](#) Sep 2026
Independent Project

- Conducted an independent strategy analysis of ASML, combining industry structure, financial performance, technology trends, and supply-chain risk to assess the durability of its competitive advantage.
- Evaluated strategic options around High-NA EUV commercialization, installed-base expansion, and selective vertical integration, developing a prioritized recommendation to sustain ASML's long-term market leadership.

Finsight [↗](#) Sep 2026 – Present
Python-based Personal Finance Analytics App

- Built a Python and Streamlit app that combines bank-statement CSVs from five bank formats plus a generic fallback, detects transfers between the user's own accounts, and runs entirely on the user's own computer.
- Developed pandas-based analytics and rule-based categorization for income, expenses, savings rate, recurring payments and monthly trends, backed by automated tests run on every push, a live online demo and a Windows desktop version.

ACTIVITIES

Japan U19 National Cricket Team — ICC Under-19 Cricket World Cup 2020
International Athlete

- Selected to represent Japan at the ICC Under-19 Cricket World Cup, competing in an international high-performance team environment.

Student Athlete

- Actively involved in the student Boxing/Kickboxing and Judo associations.

SKILLS & ADDITIONAL INFORMATION

Technical Skills: Python MATLAB Excel Siemens NX Arduino	Awards: Silver Award Scholarship (2021)
Business Skills: Cost Modelling Financial Modelling (NPV) Technology Strategy Scenario Analysis Financial / Operational Analysis	Languages: English & Hindi — Native Japanese — B2 Equivalent (JLPT N2) Dutch — Elementary (Actively learning)
Work Authorization: Eligible for Dutch Orientation Year (Zoekjaar) upon graduation.	Certifications: Corporate Finance Foundations – CFI, Sep 2026